March 27, 2026

Overview

This document describes the datasets produced by Tasks 1–3 and explains how to use them in the downstream tasks (4–8). All files are in `data/processed/`. The profit margin formula to implement in Task 4 is:

$$\boxed{\pi_{p,t,s} \;=\; P_{t,s} \;-\; C^{\text{fuel}}_{f(p),t,s} \;-\; C^{\text{O\&M}}_{f(p)} \;-\; \varepsilon_p \times P^{\text{carbon}}_{t,s}}$$

where p indexes plants, t years (2020–2050), s scenarios, and $f(p)$ the fuel type of plant p.

1 Dataset 1 — `gppd_eu_metrics.csv`

824 rows (one per EU fossil-fuel power plant) × **15 columns**.
This is the **main plant-level file**. Every row is one physical plant.

Column	Type	Unit	Description & role
`gppd_idnr`	str	—	Unique plant ID (WRI). Join key.
`name`	str	—	Plant name (display only).
`country`	str	ISO-3	Country code (DEU, POL…). **Aggregation key for Task 5.**
`country_long`	str	—	Full country name (display).
`fuel`	str	—	`coal` / `gas` / `oil`. **Aggregation key for Task 5.**
`capacity_mw`	float	MW	Installed capacity. **Weight for % stranded capacity.**
`commissioning_year`	int	year	Year of commissioning (imputed where missing). **Rule-based retirement:** stranded if year > comm_year + lifetime.
`age_2020_years`	int	years	Age of plant in 2020. Useful for vintage splits (pre/post 2000).
`lifetime`	int	years	Technical lifetime: coal=40, gas=35, oil=35.
`remaining_life_2020_years`	int	years	= lifetime − age. Negative ⇒ already past end-of-life.
`carbon_intensity_tCO2_per_MWh_elec`	float	tCO_2/MWh_{elec}	$\varepsilon_p = EF_{th}/\eta$. **Multiply by carbon price to get $C^{CO_2}_{p,t}$.**
`om_variable_eur2024_per_MWh`	float	EUR/MWh_{elec}	Variable O&M cost $C^{O\&M}$. Directly in the π formula.
`capacity_factor_2020`	float	0–0.85	Baseline capacity factor (Eurostat 2020). For utilisation stranding.
`latitude`	float	degrees	For maps.
`longitude`	float	degrees	For maps.

Key numbers by fuel:

Fuel	Plants	Capacity (GW)	ε (tCO_2/MWh)	O&M (EUR/MWh)	Avg CF 2020
Coal	279	143.3	0.920	5.0	0.48
Gas	439	142.2	0.404	4.0	0.38
Oil	106	28.9	0.762	7.0	0.15

2 Dataset 2 — `ngfs_scenarios_elec.csv`

837 rows (3 scenarios × 9 variables × 31 years) × **5 columns**.
This is the file to use for computing π. Fuel prices are **already converted to EUR2024/MWh$_{\text{elec}}$** (divided by η).

Column	Type	Values	Description
`scenario`	str	Net Zero 2050 / Delayed transition / Current Policies	NGFS scenario
`variable`	str	(see table below)	Variable name
`unit`	str	(varies)	Unit after conversion
`year`	int	2020–2050	Annual (interpolated)
`value`	float	—	Converted value

Variables and their role in the profit formula:

Variable	Unit in this file	Role in $\pi_{p,t,s}$
`Price|Carbon`	EUR2024/tCO_2	$P^{\text{carbon}}_{t,s}$ — multiply by ε_p
`Price|Secondary Energy|Electricity`	EUR2024/MWh	$P_{t,s}$ — revenue per MWh
`Price|Primary Energy|Coal`	EUR2024/MWh$_{\text{elec}}$	$C^{\text{fuel}}_{\text{coal},t,s}$ — **ready to use**
`Price|Primary Energy|Gas`	EUR2024/MWh$_{\text{elec}}$	$C^{\text{fuel}}_{\text{gas},t,s}$ — **ready to use**
`Price|Primary Energy|Oil`	EUR2024/MWh$_{\text{elec}}$	$C^{\text{fuel}}_{\text{oil},t,s}$ — **ready to use**
`Secondary Energy|Electricity|*`	MWh/yr	Generation volumes (used for CF calibration)
`Emissions|CO2|Energy|Supply|Elec.`	tCO_2/yr	Validation / sanity check only

Important: Do **not** use `ngfs_scenarios.csv` (without `_elec`). That file has fuel prices in MWh$_{\text{thermal}}$ — using it directly would **underestimate fuel costs by a factor of 2–3**×.

3 Dataset 3 — `ngfs_cf_evolution.csv`

5 580 rows (26 countries × 3 fuels × 3 scenarios × ∼24 years with nonzero capacity) × **5 columns**.

`country`	ISO-3	`fuel`	coal / gas / oil
`scenario`	str	`year`	2020–2050
`cf`	float (0–0.85)	Projected capacity factor	

Use case: Utilisation stranding robustness check (Task 7). A plant is stranded if CF $< 10\%$ for K consecutive years.

4 Dataset 4 — `gppd_eu_clean.csv`

824 rows × **10 columns.** Raw plant data before enrichment. **You should not need this file** — use `gppd_eu_metrics.csv` instead. Kept for traceability.

5 How to compute $\pi_{p,t,s}$ (Task 4)

1. **Load** `gppd_eu_metrics.csv` and `ngfs_scenarios_elec.csv`.
2. For each plant p, each year $t \in [2020, 2050]$, each scenario s:

$$
\begin{aligned}
P_{t,s} &= \texttt{Price|Secondary Energy|Electricity} \text{ at } (s,t) \\
C^{\text{fuel}}_{f(p),t,s} &= \texttt{Price|Primary Energy|[fuel]} \text{ at } (s,t) \quad (\text{already per MWh}_{\text{elec}}) \\
C^{\text{O\&M}}_{f(p)} &= \texttt{om_variable_eur2024_per_MWh} \text{ from the plant row} \\
C^{\text{CO}_2}_{p,t,s} &= \texttt{carbon_intensity_tCO2_per_MWh_elec} \times \texttt{Price|Carbon at } (s,t)
\end{aligned}
$$

3. Compute: $\pi_{p,t,s} = P_{t,s} - C^{\text{fuel}} - C^{\text{O\&M}} - C^{\text{CO}_2}$

6 How to flag stranding (Task 5)

Primary definition — Rule-based retirement: A plant p is *stranded at year* t if:

$$t > \texttt{commissioning_year}_p + \texttt{lifetime}_{f(p)}$$

Secondary definition — Economic stranding: A plant p is *stranded at year* t under scenario s if:

$$\pi_{p,\tau,s} < 0 \quad \text{for } K \text{ consecutive years ending at } \tau = t \quad (K = 3 \text{ recommended})$$

Robustness — Utilisation stranding: Using `ngfs_cf_evolution.csv`: a plant is stranded if $\text{CF}_{c,f,t,s} < 0.10$ for $K = 3$ consecutive years.

7 How to aggregate (Task 5)

For each country c, fuel f, scenario s, year t:

$$\%\ \text{Stranded capacity}_{c,f,t,s} = \frac{\sum_{p \in (c,f)} \mathbf{1}\{\text{stranded}_{p,t,s}\} \times \texttt{capacity_mw}_p}{\sum_{p \in (c,f)} \texttt{capacity_mw}_p}$$

Report by country, by fuel, and by vintage (pre/post 2000 using `age_2020_years` 20).

8 Three scenarios at a glance

	Net Zero 2050	**Delayed Transition**	**Current Policies**
Type	Orderly	Disorderly	Insufficient
Carbon price 2020	37 EUR/tCO_2	37 EUR/tCO_2	37 EUR/tCO_2
Carbon price 2030	∼377	∼37 (flat)	∼37 (flat)
Carbon price 2050	∼1 386	∼954	∼37 (flat)
Elec. price 2050	∼58 EUR/MWh	∼62 EUR/MWh	∼66 EUR/MWh
Coal gen. 2050	≈ 0	≈ 0	≈ 0
Key pattern	Progressive squeeze	Shock post-2030	Minimal transition

9 Unit conversion summary

All monetary values are in **EUR2024**. Conversion from NGFS raw USD2010:

$$\text{Factor} = \underbrace{1.438}_{\text{CPI inflation}} \times \underbrace{0.924}_{\text{USD}\to\text{EUR}} = 1.33$$

Sources: BLS CPI-U annual averages (218.056 → 313.515); ECB reference rate 2024 (1 EUR = 1.0822 USD).

10 Known limitations to mention in the report

- NGFS region is EU28 (includes UK), used as proxy for EU27.
- Single IAM model (REMIND-MAgPIE 3.3-4.8). The project brief suggests comparing several.

- 36.7% of commissioning years were imputed (median by country $\times$ fuel).
- Oil trajectories are identical across all 3 scenarios (oil is residual in EU mix).
- Some country$\times$fuel pairs have $CF_{baseline}$=0 (plants in reserve, no generation in 2020).